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# ============================================
# Lab: Bag of Words Implementation
# Name: Your Name
# Roll No: Your Roll Number
# ============================================

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# STEP 1: Import Libraries
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# For text processing
import nltk
import string

# For Bag of Words
from sklearn.feature_extraction.text import CountVectorizer

# Download stopwords (only first time)
nltk.download('stopwords')

from nltk.corpus import stopwords

print("Libraries Loaded Successfully!\n")

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# STEP 2: Sample Dataset
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documents = [
    "I love natural language processing",
    "Natural language processing is very interesting",
    "I love machine learning",
    "Machine learning and natural language processing are related"
]

print("Original Documents:\n")
for doc in documents:
    print(doc)

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# STEP 3: Text Preprocessing
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stop_words = set(stopwords.words('english'))

processed_docs = []

for doc in documents:
    # Convert to lowercase
    doc = doc.lower()

    # Remove punctuation
    doc = doc.translate(str.maketrans('', '', string.punctuation))

    # Remove stopwords
    words = doc.split()
    words = [word for word in words if word not in stop_words]

    processed_docs.append(" ".join(words))

print("\nProcessed Documents:\n")
for doc in processed_docs:
    print(doc)

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# STEP 4: Create Bag of Words Model
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vectorizer = CountVectorizer()

bow_matrix = vectorizer.fit_transform(processed_docs)

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# STEP 5: Display Vocabulary
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print("\nVocabulary:\n")
print(vectorizer.get_feature_names_out())

# -------------------------------
# STEP 6: Display Bag of Words Matrix
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print("\nBag of Words Matrix (Dense Form):\n")
print(bow_matrix.toarray())

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# STEP 7: Show Word Counts per Document
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feature_names = vectorizer.get_feature_names_out()
matrix = bow_matrix.toarray()

print("\nWord Counts per Document:\n")

for i, row in enumerate(matrix):
    print(f"\nDocument {i+1}:")
    for word, count in zip(feature_names, row):
        if count > 0:
            print(f"{word} : {count}")

print("\nBag of Words Model Completed Successfully!")
```

```
Libraries Loaded Successfully!

Original Documents:

I love natural language processing
Natural language processing is very interesting
I love machine learning
Machine learning and natural language processing are related

Processed Documents:

love natural language processing
natural language processing interesting
love machine learning
machine learning natural language processing related

Vocabulary:

['interesting' 'language' 'learning' 'love' 'machine' 'natural'
 'processing' 'related']

Bag of Words Matrix (Dense Form):

[[0 1 0 1 0 1 1 0]
 [1 1 0 0 0 1 1 0]
 [0 0 1 1 1 0 0 0]
 [0 1 1 0 1 1 1 1]]

Word Counts per Document:


Document 1:
language : 1
love : 1
natural : 1
processing : 1

Document 2:
interesting : 1
language : 1
natural : 1
processing : 1

Document 3:
learning : 1
love : 1
machine : 1

Document 4:
language : 1
learning : 1
machine : 1
natural : 1
```

```
processing : 1
related : 1

Bag of Words Model Completed Successfully!
[nltk_data] Downloading package stopwords to /root/nltk_data...
```